Liu Chee Wei (Jacky)

R&D Software Manager · PTEM

ENGINEERING LEADERSHIP · MANUFACTURING TEST · OPENTAP · INSTRUMENT AUTOMATION

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PROFILE

R&D Software Manager at Keysight Technologies, leading a **14-engineer team** that designs and builds **PathWave Test Executive for Manufacturing (PTEM)**, a manufacturing test executive used globally. The team extends OpenTAP with platform features, instrument plugins, and test solutions for automotive and manufacturing applications. Previously PTEM's lead engineer and architect (2019–2026). **15 years** in test and automation software; **co-inventor of two US patents** in automated LiDAR target simulation. Hands-on in C#, C++ and Python.

14	**PTEM**	**2**	**15 yrs**
engineers led on PTEM	manufacturing test executive, used globally	US patents, LiDAR target simulation	test & automation software

EXPERIENCE

Keysight Technologies — Penang, Malaysia

R&D Software Manager — Sep 2026 – Present

- Lead a **14-engineer R&D team** designing and building **PTEM** (PathWave Test Executive for Manufacturing), used by customers globally.
- Drive the PTEM roadmap on OpenTAP: user-facing platform features, instrument plugins, and test solutions for automotive and manufacturing applications.
- Hire, coach and develop engineers; own delivery planning and stakeholder alignment across global programs.

R&D Software Lead Engineer — Oct 2019 – Aug 2026

- Led a team of 10 on **PTEM architecture and development** on OpenTAP, from platform features to instrument plugins.
- Scrum Master; **established the SQ release process** across global programs.
- Sole software developer on the LiDAR Target Simulator project; **co-invented 2 US patents** in automated LiDAR target simulation scanning.

WAFTECH — Penang, Malaysia

Motion Control & Software Dev Senior Engineer — Apr 2016 – Sep 2019

- Built next-generation machine control software in C# and WPF.
- Implemented SECS/GEM integration for semiconductor equipment communication.
- Led on-site deployments and mentored junior engineers.

Machine Automation Software Engineer — Oct 2014 – Apr 2015

- Developed motion control and wafer handling software in C# and WPF.

Makerzone Sdn Bhd — Penang, Malaysia

Co-Founder — Apr 2011 – Mar 2016

- Co-founded and ran a 3D printing service business; built the company website and delivered customer projects from design to delivery.

Keysight Technologies — Penang, Malaysia

R&D System Engineer — Jun 2011 – Oct 2014

- Developed automotive test systems in C++; supported the TS-5000 hardware handler library.

SKILLS

Leadership	Engineering management, hiring & coaching, roadmapping, Scrum Master, release process
Platforms	PTEM, OpenTAP, .NET, WPF, Git, Claude API
Languages	C#, C/C++, Python, JavaScript, PowerShell
Domain	Manufacturing test, automotive test, instrument automation, LiDAR/ADAS, SECS/GEM
Spoken	Chinese (native), English (professional), Malay (conversational)

PATENTS

US2022260714A1
Automated LiDAR Target Simulation Scanning Systems & Methods

US20230132855A1
Automated LiDAR Target Simulation Scanning Systems and Methods

EDUCATION & CERTIFICATIONS

B.Eng. (Hons) Electrical & Electronic Eng. 2007 – 2011
Universiti Tunku Abdul Rahman (UTAR)

Agile SCRUM for Practitioners Feb 2026
DreamCatcher Consulting

Practical AI with Machine & Deep Learning Mar 2021
DreamCatcher Consulting